PROBLEM STATEMENT

ContextGuard: An LLM-Based Semantic Data Quality Auditor for ML Pipelines

DL · Generative AI · NLP · Tabular Data

DSAI Lab Project · IIT Madras BS in Data Science and Applications · Group 1

Machine-learning models are only as reliable as the data they are trained on, yet real-world tables often contain errors that conventional data-quality checks miss. This project targets cell-level errors in single relational tables, in four categories: missing or placeholder values, format and pattern inconsistencies, violations of dependencies between columns (for example, a city that does not match its zip code), and contextual outliers (values that are plausible in isolation but wrong given the rest of the row). Rule-based tools require experts to hand-write constraints for every dataset. Statistical detectors judge each value in isolation, so they flag rare-but-correct values and miss plausible-looking wrong ones. Many of these errors can only be resolved by reasoning over the surrounding row and table context, which is where we expect a language model to add value.

> We propose a hybrid pipeline. Lightweight statistical and rule-based detectors first identify candidate cells with high recall. A language model then examines each candidate together with its row, the column's profile and dependency evidence from the table. It decides whether the cell is genuinely erroneous, identifies the error type, proposes a correction and assigns a confidence score. Only corrections above a calibrated confidence threshold are applied automatically; uncertain cases are routed for human review. We will start with zero-shot and few-shot prompting of an open model in the 3–8B parameter range (such as Qwen3-4B). Parameter-efficient fine-tuning (QLoRA) will be adopted if preliminary experiments show a clear gap. The final choice of model, fine-tuning strategy and architecture will be justified through these pilot experiments.

> The TableEG benchmark is our primary dataset. It provides real-world tables from several domains, each with clean and dirty versions and annotated errors. Additional or alternative datasets may be incorporated based on the literature review. To test generalisation, some tables will be held out entirely and used only for evaluation.

Evaluation will cover the detection stage and the complete pipeline separately, so that errors missed by the initial detectors are measured rather than hidden. Metrics include error-detection precision, recall and F1, correction accuracy, and the rate at which originally correct values are wrongly modified. Performance will be reported per error category, together with an analysis of confidence-threshold reliability and of how much human review is needed. The approach will be compared against established baselines, including statistical and rule-based detectors and zero-shot LLM prompting, with the full baseline set finalised in Milestone 1. We will also assess downstream impact by comparing a model trained on uncleaned versus cleaned data, report computational cost and inference latency, and, where feasible, run ablations to isolate the contributions of statistical detection, contextual reasoning and fine-tuning.

The project will be considered successful if it shows a clear, measurable improvement over the baselines on unseen tables without a meaningful increase in wrongly modified values. Specific targets will be set after the pilot experiments. As a final deliverable, the evaluated system will be packaged as an application where a user uploads a dataset and receives a cleaned version, along with a log of the changes made and the cells flagged for review. The application demonstrates the solution; the experimental results remain the principal measure of success.